

$$X_{2j} = X_{3j} + X_{24} \quad j = 5, \dots, n$$

$$X_{3j} = X_{1j} - X_{14} \quad j = 5, \dots, n-1$$

$$X_{3n} = -X_{14}$$

$$X_{2j+1} = X_{3j+1} + X_{24}$$

$$= X_{1j+1} - X_{14} + X_{24}$$

suppose $X_{24} \rightarrow \infty$ $\begin{cases} X_{35} \rightarrow \infty \\ X_{15} \text{ finite} \end{cases}$ and choose X_{25} finite

$\begin{cases} X_{37} \rightarrow \infty \\ X_{17} \text{ finite} \end{cases}$ and choose X_{27} finite

$$X_{25} = X_{35} + X_{24} = X_{15} - X_{14} + X_{24}$$

$$X_{27} = X_{37} + X_{24} = X_{17} - X_{14} + X_{24}$$

$$\rightarrow X_{27} = X_{25} - X_{15} + X_{17}$$

you can only pick one X_{2j} to be finite

what if both X_{35} and X_{15} are ∞ ?

X_{36} and X_{16} are ∞

could have X_{24} and X_{25} and X_{26} finite

$$X_{26} = X_{25} - X_{15} + X_{16}$$

back to 1-zero

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{15} - X_{13}$$

$$X_{26} = -X_{13}$$

rank: 3

6 variables (13, 14, 15, 24, 25, 26)

our trick is to choose some to be independent and send rest to ∞

• can't have 13 finite otherwise 26 depends on it but finite ($13 \rightarrow \infty$)

• if X_{14} is chosen as indep. and finite then $X_{24} \rightarrow \infty$

• if X_{24} is chosen as indep. and finite then $X_{14} \rightarrow \infty$

2 - 2 revs

$$\underline{X_{35}} = X_{15} - \underline{X_{14}}$$

$$\underline{X_{36}} = -\underline{X_{14}}$$

$$X_{25} = \underline{X_{35}} + X_{24}$$

$$X_{26} = \underline{X_{36}} + X_{24}$$

rank: 4
total variables: 7 $\rightarrow$ at most 3 independent

$$X_{14} = \lambda$$

$$X_{36} = -\lambda$$

$$X_{35} = X_{15} - \lambda$$

$$X_{25} = X_{15} - \lambda + X_{24}$$

$$X_{26} = -\lambda + X_{24}$$

$$\rightarrow \begin{cases} X_{35} = X_{15} - \lambda \\ X_{26} = X_{24} - \lambda \\ X_{25} = X_{15} + X_{26} \end{cases}$$

setting 14, 36 $\rightarrow \infty$ leaves 5 $\rightarrow$ at least 2 more have to go to ∞

or 25, 26 be independent?

• if 2 variables are in the same eq they cannot both be independent.

$\rightarrow$ at most one of the X_{2j} is independent.

$$\begin{cases} X_{35} = X_{15} - \lambda \\ X_{26} = X_{24} - \lambda \\ X_{25} = X_{15} + X_{26} \end{cases} \rightarrow \begin{cases} X_{35} = X_{15} - \lambda \\ X_{35} - X_{15} = X_{26} - X_{24} \\ X_{25} = X_{15} + X_{26} \end{cases}$$

• X_{24} finite: $X_{26} \rightarrow -\infty$.

• if X_{15} finite then $X_{25}, X_{35} \rightarrow \infty$

~~• if X_{35} finite then $X_{15} \rightarrow \infty$ and $X_{25} = X_{35} + \lambda + X_{24} - \lambda = X_{35} + X_{24}$ No!~~

• X_{25} finite:

~~• if X_{35} finite: $X_{25} = X_{35} + X_{24}$ No!~~

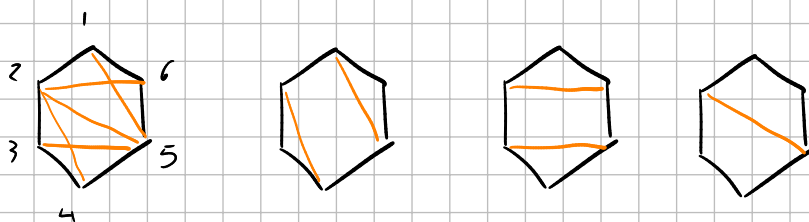
~~• if X_{15} finite: $X_{25} = X_{15} + X_{26}$ No!~~

• X_{26} finite:

~~• if X_{15} finite: $X_{25} = X_{15} + X_{26}$ No!~~

• if X_{35} finite: $X_{15} \rightarrow \infty, X_{25} \rightarrow \infty$

choices: (24, 15) (25) (26, 35) as finite and independent



$n=7$

$$X_{35} = X_{15} - X_{14}$$

$$X_{36} = X_{16} - X_{14}$$

$$X_{37} = -X_{14}$$

$$X_{25} = X_{35} + X_{24}$$

$$X_{26} = X_{36} + X_{24}$$

$$X_{27} = X_{37} + X_{24}$$

$$X_{3j} = X_{31} + X_{1j}$$

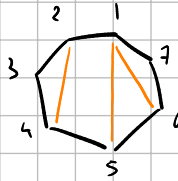
$$X_{2j} = X_{1j} + X_{27} \quad j=4, \dots, 6$$

$$X_{2j} = X_{3j} + X_{24} \quad j=5, \dots, 7$$

• if X_{24} finite:

not have $X_{35}, X_{36} \rightarrow \infty$ and $X_{25}, X_{26}, X_{27} \rightarrow \infty$

can choose X_{15}, X_{16} finite.



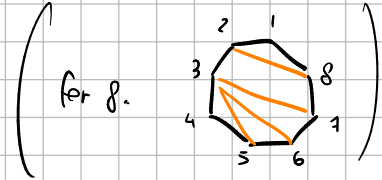
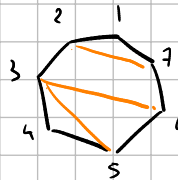
$$X_{2j} - X_{3j} = X_{27} - X_{37}$$

$$X_{2j} - X_{1j} = X_{27} - X_{17}$$

• if X_{27} finite: (symmetric picture)

not have $X_{15}, X_{16} \rightarrow \infty$ and $X_{25}, X_{26}, X_{24} \rightarrow \infty$

can choose X_{35}, X_{36} finite.



• if X_{25} finite:

• not choose $X_{24}, X_{35} \rightarrow \infty$

• not choose $X_{27}, X_{15} \rightarrow \infty$

• if X_{16} finite: $X_{36} \rightarrow \infty$

$$X_{26} = X_{16} + X_{25} - X_{15} \quad \text{so} \quad X_{26} \rightarrow \infty$$

• if X_{36} finite: $X_{16} \rightarrow \infty$

$$X_{26} = X_{36} - X_{35} + X_{25} \rightarrow \infty$$

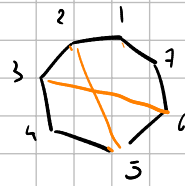
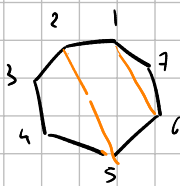
$$X_{26} = X_{16} + X_{27}$$

$$X_{26} = X_{36} + X_{24}$$

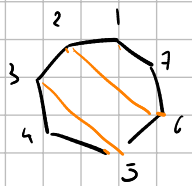
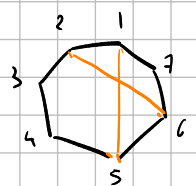
$$X_{25} = X_{15} + X_{27}$$

$$X_{25} = X_{35} + X_{24}$$

$\Rightarrow$ clearly X_{25} finite forces $X_{15}, X_{35}, X_{24}, X_{26}, X_{27} \rightarrow \infty$



for 26: $X_{16}, X_{36} \rightarrow \infty \quad X_{24}, X_{26}, X_{27} \rightarrow \infty$



$$X_{35} = X_{15} - X_{14}$$

$$X_{36} = X_{16} - X_{14}$$

$$X_{25} = X_{35} + X_{24}$$

$$X_{26} = X_{36} + X_{24}$$

$$X_{27} = X_{37} + X_{24}$$

25, 16 free

$$X_{25} = x \quad X_{16} = y$$

$$X_{14} = \lambda \quad X_{37} = -\lambda$$

$$X_{24} = -\lambda$$

$$X_{35} = X_{15} - \lambda$$

$$x = X_{15} - 2\lambda$$

$$X_{36} = y - \lambda$$

$$X_{26} = y - 2\lambda$$

$$X_{27} = -2\lambda$$

$$X_{15} = x + 2\lambda \rightarrow +\infty$$

$$X_{35} = x + \lambda \rightarrow +\infty$$

$$X_{36} = y - \lambda \rightarrow -\infty$$

✓

25, 36 free

$$X_{25} = x \quad X_{36} = y$$

$$X_{14} = \lambda \quad X_{37} = -\lambda$$

$$X_{35} = X_{15} - \lambda$$

$$x = X_{15} + X_{27}$$

$$y = X_{16} - \lambda$$

$$X_{26} = y + X_{24}$$

$$X_{27} = -\lambda + X_{24}$$

if we choose

$$X_{24} = -\lambda \rightarrow -\infty$$

$$X_{26} = y - \lambda \rightarrow -\infty$$

$$X_{15} = x + 2\lambda \rightarrow \infty$$

$$X_{27} = -2\lambda \rightarrow -\infty$$

$$X_{35} = x + \lambda \rightarrow \infty$$

$$\begin{cases} X_{26} = y - \lambda \\ X_{15} = x + 2\lambda \\ X_{35} = x + \lambda \end{cases}$$

generic n:

pick 25: can choose one from each pair of (16, 36), (17, 37), ..., (1n-1, 3n-1)

Ⓐ: if we pick 24 free: we can only pick 15, 16, ..., 1n-1 free

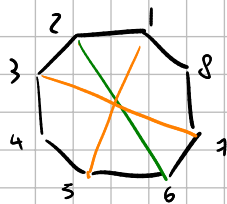
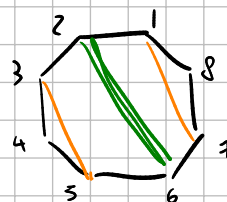
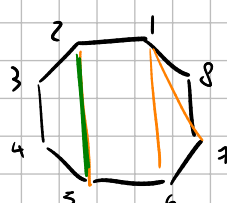
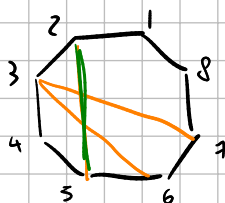
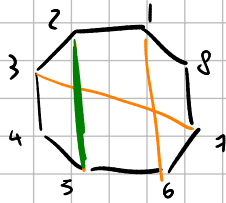
Ⓑ: if we pick 27 free: we can only pick 35, 36, ..., 3n-1 free

Ⓒ: if we pick 2j j=5, ..., n-1: we can pick one from each pair (1l, 3l) with $l \in \{5, \dots, n-1\}$ and $l \neq j$

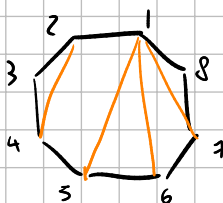
→ to be verified → not true!

Ⓓ: if we pick no 2j then can pick one from each pair (1l, 3l), $l=5, \dots, n-1$

⑦

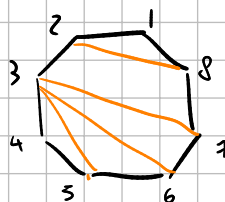


⑧

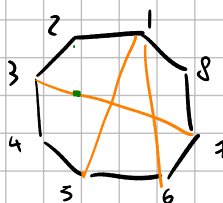


1 case

⑨



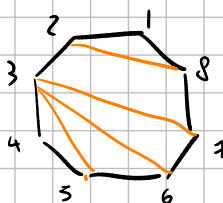
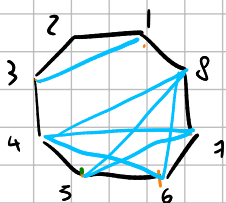
⑩



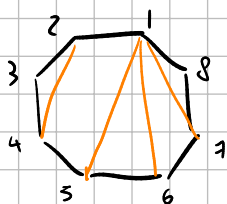
which are free?

13

46 47 48
57 58
68



⑪



⑫

$$n = 8$$

$$k = 3$$

$$X_{46} = X_{16} - X_{15}$$

$$X_{47} = X_{17} - X_{15}$$

$$X_{48} = -X_{15}$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{47} = X_{27} - X_{25}$$

$$X_{48} = X_{28} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{47} = X_{37} - X_{35}$$

$$X_{48} = X_{38} - X_{35}$$

$$X_{15} = \lambda \Rightarrow \infty$$

$$X_{48} = -\lambda \Rightarrow -\infty$$

$$X_{46} = X_{16} - \lambda$$

$$X_{47} = X_{17} - \lambda$$

$$X_{46} = X_{26} - X_{25}$$

$$X_{47} = X_{27} - X_{25}$$

$$-\lambda = X_{28} - X_{25}$$

$$X_{46} = X_{36} - X_{35}$$

$$X_{47} = X_{37} - X_{35}$$

$$-\lambda = X_{38} - X_{35}$$

$$X_{16} = X_{26} - X_{28}$$

$$X_{17} = X_{27} - X_{28}$$

$$X_{16} = X_{36} - X_{38}$$

$$X_{17} = X_{37} - X_{38}$$

• if X_{25} finite:

$$28 \rightarrow \infty$$

$$27, 47 \rightarrow \infty$$

$$26, 46 \rightarrow \infty$$

or only choose 16, 17 as finite

• if X_{35} finite:

$$38 \rightarrow \infty$$

$$37, 47 \rightarrow \infty$$

$$36, 46 \rightarrow \infty$$

or only choose 16, 17 as finite

$$X_{24} = X_{14} - X_{13}$$

$$X_{25} = X_{15} - X_{13}$$

$$X_{26} = -X_{13}$$

$$X_{24} + X_{25} = X_{14} + X_{15} - 2X_{13}$$

$$X_{14} = 2$$

$$X_{15} = 6$$

$$X_{13} = \lambda$$

$$X_{26} = -\lambda$$

$$X_{24} = 2 - \lambda$$

$$X_{25} = 6 - \lambda$$

but one of the "bad ones"

$$n=8$$

$$X_{35} = X_{15} - X_{14}$$

$$X_{36} = X_{16} - X_{14}$$

$$X_{37} = X_{17} - X_{14}$$

$$X_{38} = -X_{14}$$

$$X_{25} = X_{35} + X_{24}$$

$$X_{26} = X_{36} + X_{24} \quad \textcircled{*}$$

$$X_{27} = X_{37} + X_{24} \quad \textcircled{**}$$

$$X_{28} = X_{38} + X_{24}$$

choose X_{26} and X_{27} as finite

$$24 \rightarrow \infty$$

$$37 \rightarrow -\infty$$

$$\text{if } 17 \text{ finite: } X_{27} = X_{17} + X_{28} \quad \text{so } |28| \rightarrow \infty \quad |17| \rightarrow \infty$$

use

$\textcircled{*}$
 $\textcircled{**}$

together:

$$X_{26} - X_{36} = X_{27} - X_{37}$$

not allowed! we already used these eqs

test

$$X_{35} = X_{15} - \lambda$$

$$X_{36} = X_{16} - \lambda$$

$$X_{37} = X_{17} - \lambda$$

$$X_{25} = X_{35} + X_{24}$$

$$X_{26} = X_{36} + X_{24}$$

$$X_{27} = X_{37} + X_{24}$$

$$X_{14} = \lambda$$

$$X_{38} = -\lambda$$

$$X_{24} = -\lambda$$

$$X_{28} = -2\lambda$$

$$X_{15} = x$$

$$X_{26} = y$$

$$X_{27} = z$$

$$y = X_{36} - \lambda$$

$$z = X_{37} - \lambda$$

$$X_{36} = X_{16} - \lambda$$

$$X_{37} = X_{17} - \lambda$$

$$X_{35} = x - \lambda \rightarrow \infty$$

$$X_{25} = x - 2\lambda \rightarrow \infty$$

$$X_{16} = y + 2\lambda \rightarrow \infty$$

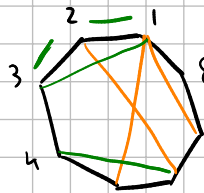
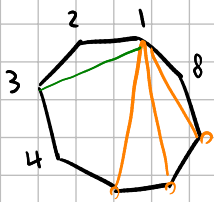
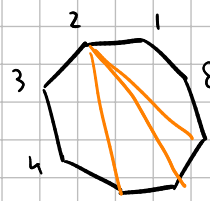
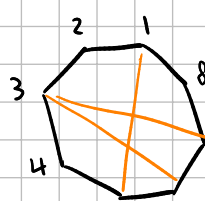
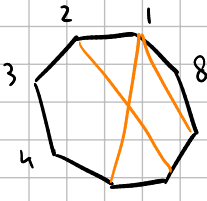
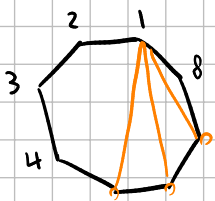
$$X_{17} = z + 2\lambda \rightarrow \infty$$

$$X_{36} = y + \lambda \rightarrow \infty$$

$$X_{37} = z + \lambda \rightarrow \infty$$

so rule (c) seem to be:

for each triple $(1j, 2j, 3j)$ $j=5, \dots, n-1$ we pick one.



→ for edges 12 13 earlier noted for 2-211 didn't work.

